

DATA STORYTELLING REPORT

Online Retail Transaction Analysis

Internship Task 2 • Submitted by Rahul Kumar
Decision-oriented analysis built from the completed EDA work

Executive Summary

This report converts the completed Online Retail exploratory analysis into a decision-oriented data story. The workflow moves from data quality and preparation to business metrics, relationship analysis, key findings, limitations and practical recommendations. The objective is not only to describe the dataset, but to explain what the observed patterns could mean for sales planning and customer-focused decisions.

1. Objective and Business Questions

The analysis answers four practical questions: (1) What is the quality of the transaction data? (2) What is the overall sales scale? (3) Which markets/products and relationships stand out? (4) What actions are reasonable given the evidence?

2. Dataset and Legal Data Use

The project uses the UCI Online Retail dataset. The UCI record describes transaction fields such as invoice, product, quantity, invoice date, unit price, customer ID and country, and lists the dataset under a CC BY 4.0 license. The project README instructs the user to obtain the dataset from the official UCI source rather than bundling the raw Excel file.

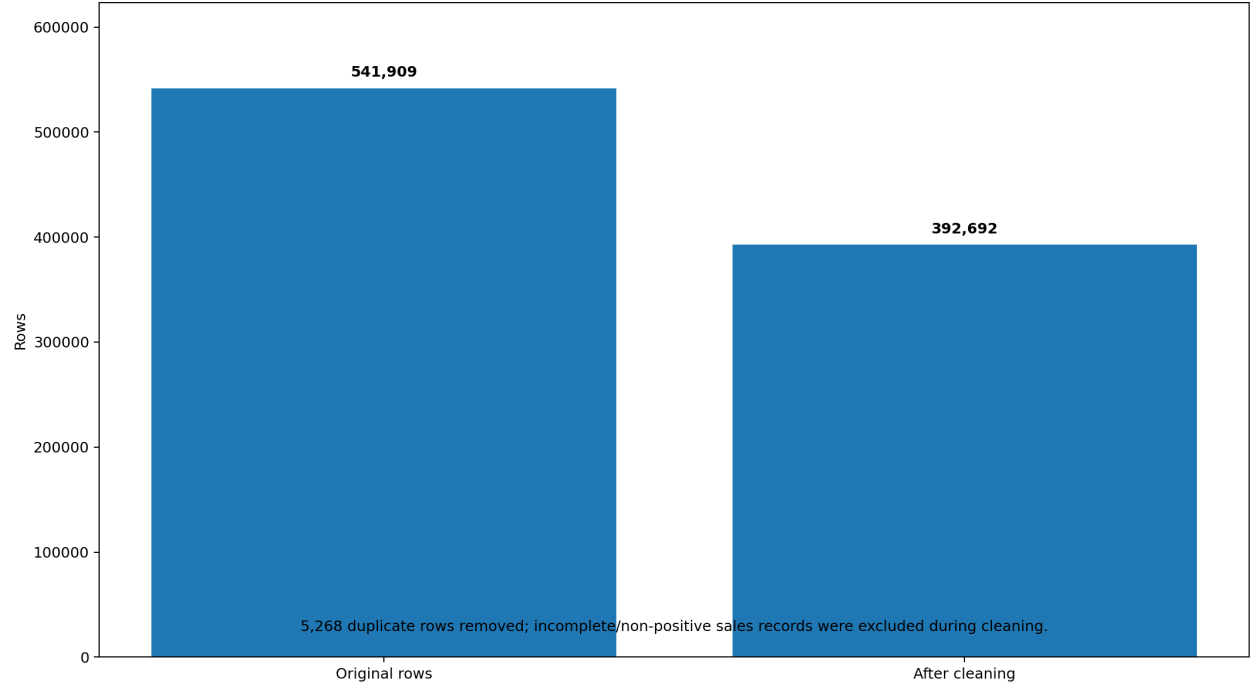
3. Methodology

The pipeline loads the Excel data, removes exact duplicate rows, removes records without required product/customer information, converts dates and numeric fields, keeps positive quantity and unit-price rows for sales-performance analysis, creates Revenue = Quantity × UnitPrice, and derives calendar features. Aggregations are then used for revenue, orders, customers, products, country/product rankings and time-based analysis.

4. Data Quality Notes

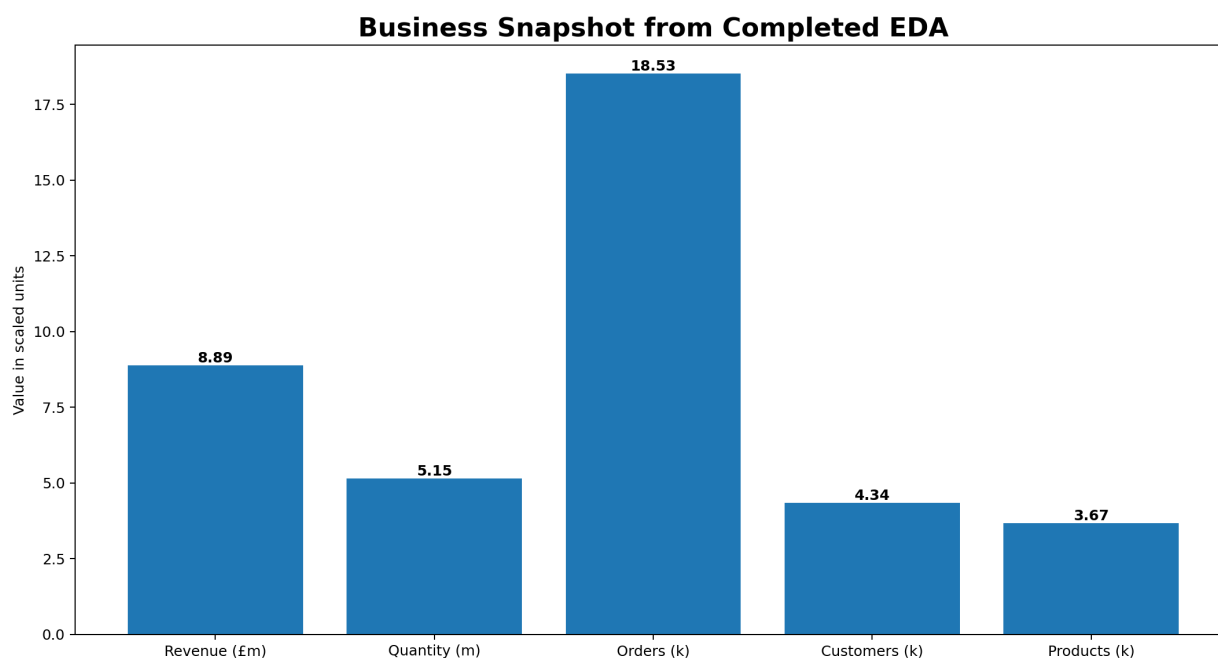
Measure	Result
Original rows	541,909
Original columns	8
Duplicate rows removed	5,268
Cleaned rows	392,692
Cleaned columns	14
Missing values after cleaning	0

Data Quality: Analytical Dataset Reduction



5. Business Metrics

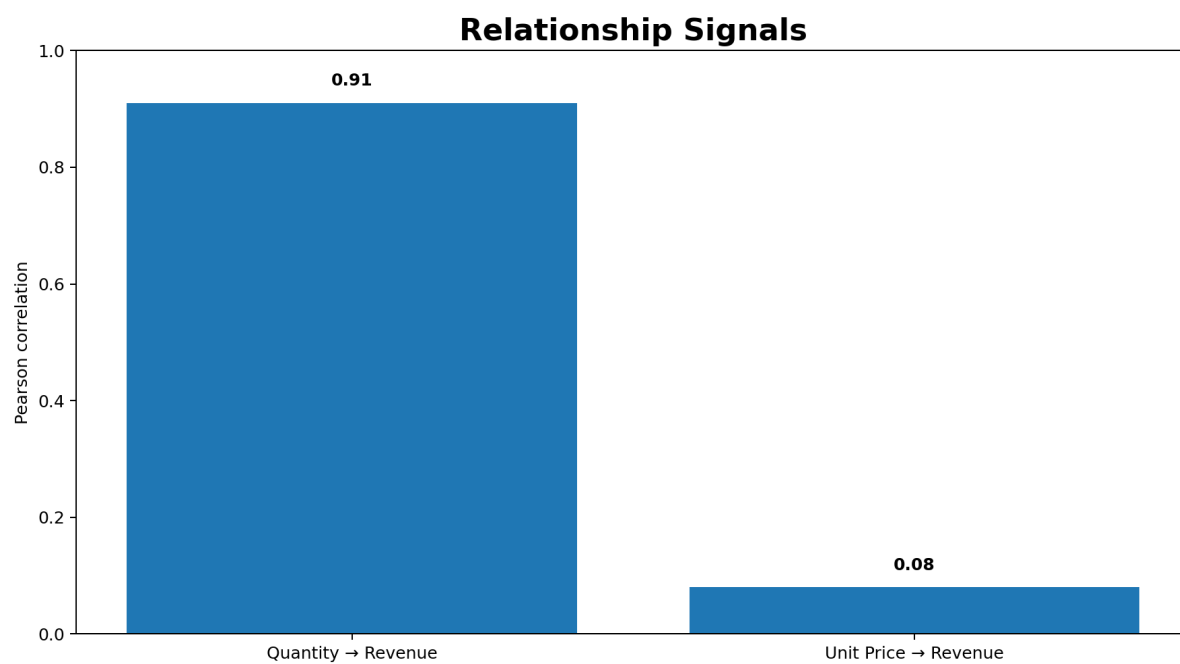
Metric	Result
Total revenue	£8,887,208.89
Total quantity sold	5,152,002
Total orders	18,532
Total customers	4,338
Total products	3,665
Average order value	£479.56



Revenue, quantity, orders, customers and products use different units; values are shown on scaled axes for communication only.

6. Key Findings

- The United Kingdom is the dominant revenue-generating country in the analyzed sales view.
- Revenue increased strongly during September–November 2011, with November recording the highest monthly revenue.
- PAPER CRAFT, LITTLE BIRDIE is the highest-revenue product in the completed analysis.
- Quantity and Revenue have a strong positive correlation of approximately 0.91.
- UnitPrice and Revenue have a very weak linear correlation of approximately 0.08.
- Revenue is highly right-skewed, indicating that a relatively small number of transactions contribute very large revenue values.



Correlation indicates association, not causation.

7. Decision-Oriented Interpretation

Market focus: Because the UK is the largest revenue contributor, customer retention, repeat-purchase programs and market-specific merchandising can have high impact. This is a prioritization signal, not evidence that other countries should be ignored.

Seasonal planning: The September–November rise, especially the November peak, supports earlier inventory planning and campaign preparation for the strongest observed period.

Product availability: The leading product by revenue deserves availability monitoring and cross-sell analysis. A single product's historical leadership does not guarantee future demand.

Volume strategy: The strong Quantity–Revenue relationship suggests that basket size, repeat purchases and volume-oriented merchandising deserve attention. The weak UnitPrice–Revenue relationship means price alone should not be treated as the primary revenue driver from this analysis.

8. Actionable Recommendations

1. Build a UK-focused retention dashboard tracking repeat customers, order frequency and customer value.
2. Use the September–November period as a planning window for stock, promotions and campaign readiness.
3. Monitor the highest-revenue products for stock-out risk and test related-product bundles.
4. Segment customers by purchase frequency and order value before designing targeted offers.
5. Re-run the pipeline periodically so recommendations reflect current transactions rather than historical data alone.

9. Limitations

The cleaned sales-performance view excludes records with missing Description or CustomerID and excludes non-positive Quantity or UnitPrice values. This can remove cancellations, returns and incomplete customer records from the decision view. The dataset is historical, so it cannot by itself establish future demand. Correlation measures association rather than causation. Results also depend on the chosen cleaning rules and aggregation definitions.

10. Tools and External Resources

Python, pandas, NumPy, Matplotlib, Seaborn, OpenPyXL and ReportLab were used. The primary dataset is the UCI Online Retail dataset. Pandas documentation, Matplotlib documentation and Kaggle Learn were used as learning/reference resources. No APIs, models, images, passwords, tokens or private credentials are required for the analysis.

11. Conclusion

The project turns a large transaction table into a compact decision story. The strongest signals are the dominance of the UK market, the late-2011 revenue peak, the leading product, and the strong association between quantity and revenue. These findings support practical actions around retention, seasonal planning, availability and basket-building, while the limitations section defines where further analysis is required before making high-stakes business decisions.

GitHub repository from the completed Task 1 project: github.com/rahulraj2007k-pixel/EDA-Dashboard